

I — Contributions to research and development

a. Articles published or accepted in peer-reviewed journals

Wolkovich, E.M., **Rouleau-Desrochers, C.**, García de Cortázar-Atauri, I., Walker, M.A., & Lacombe, T. (2025) Building a more predictive model of terroir for the Anthropocene. *Plants, People, Planet*. 7(6): 1611–1620. DOI: 10.1002/ppp3.70055 (BSc work).

I developed the meta-analysis conceptual figure and led the global mapping analysis identifying key data gaps; I assisted with the writing and edited the methods and implications sections.

b. Other peer-reviewed contributions

Rouleau-Desrochers, C.*, Van der Meersch, V., Pederson, N., & Wolkovich, E.M. (2026) Evidence that growing season length and tree growth are decoupled in an urban arboretum. *Canadian Society for Ecology and Evolution (CSEE)*. Full oral presentation at a national-level conference (MSc work).

Rouleau-Desrochers, C.*, Pederson, N., & Wolkovich, E.M. (2025) How growing season length affects tree growth in a common garden study. *EcoEvo*. Oral presentation, regional-level conference (MSc work).

c. Non-peer-reviewed contributions

Rouleau-Desrochers, C., Van der Meersch, V., Buonaiuto, D.M., Chamberlain, C.J., Loughnan, D., Pederson, N., & Wolkovich, E.M. (2026) Warmer and longer growing seasons increase growth for juvenile trees in a common garden. *Global Change Biology*. Submitted September 2026, 28 pp. (MSc work).

I developed the questions, collected and analysed the data, led the writing and the interpretation of the results and their implications for forest regeneration and tree responses to climate change.

Rouleau-Desrochers, C., Patry, É., Gingras, G., Ucler, J., Vézina-Doré, M., Paillé, O., & Moghaddam, Y.Y. (2024) “Un été parti en fumée.” *Le Point Bio*. 18: 9–19 (BSc work).

As lead author, I conducted the interviews with researchers and wrote the manuscript.

II — Most significant contributions to research and development

1. Warmer and longer growing seasons increase growth for juvenile trees in a common garden (Global Change Biology and CSEE 2026). Extended growing seasons are widely assumed to increase tree growth, yet recent work repeatedly fails to find this link, largely because it focused on early-season leafout, when cool temperatures limit photosynthesis. I designed and led an analysis testing how juvenile tree growth in four Betulaceae species responds to both season length and thermal sum, using start- and end-of-season phenology data across four provenances. I found that warmer seasons were a better predictor of growth than season length in most species, and that end-of-season phenology was a stronger predictor than start-of-season, challenging assumptions about when in the season growth actually happens. I also tested for provenance and found little evidence of local adaptation, suggesting these responses may generalize across species ranges. I wrote the manuscript and led the interpretation; my co-authors advised on the modelling and on the tree-ring data collected at the Arnold Arboretum. We targeted *Global Change Biology* for its readership of ecologists and modellers working on climate-driven shifts in ecosystem processes. My work shows that season-length effects are species- and timing-dependent, with direct implications for how carbon models incorporate phenology.

2. Predictive model of terroir under climate change (Plants, People, Planet, 2025). Current vine-planting choices and sparse genotype $\times$ environment data undermine adaptation planning under climate change. I synthesized fragmented genotype $\times$ environment datasets into a proposed meta-analysis frame-

work and designed a global data-gap map that reframed the paper’s core argument: predictive models must explicitly account for missing data to guide adaptation. Working with collaborators in France and the United States, I shaped the manuscript’s conceptual structure and contributed to the writing. We chose *Plants, People, Planet* for its mandate to bridge fundamental plant science and applied practitioner audiences, matching our goal of making the findings usable for viticulture planning.

III — Applicant’s statement

Research experience: I first fell into natural-science research in 2021 with Dr. Pierre Drapeau, investigating woodpecker damage to electrical infrastructure across Québec, where I learned rigorous field protocols, database management and scientific coding. A 2022 Undergraduate Student Research Award (USRA) took me to remote boreal forest for fieldwork on Pileated Woodpecker habitat—banding, nest-searching and tree-classification—and I leveraged 20 years of nesting data to test how their nesting phenology is shifting with climate change, my first independent statistical project. A second USRA at UBC in 2023 (supervised by Drs. Frederik Baumgarten and E.M. Wolkovich) introduced me to large-scale experiments: over that summer I designed and implemented a defoliation experiment to test tree resilience, maintained a 1,000-tree trial, deployed dendrometers and analysed growth responses to treatments.

In 2024, I conceived and led a seven-species greenhouse experiment on growing season length, formulating the hypotheses, designing the treatments and integrating climate and phenological data into statistical models. Alongside this, I joined field campaigns in Smithers (BC) and Mount Rainier (WA) to study forest regeneration dynamics, where I improved my forestry methods and learned the basics of dendrochronology. In 2025, while continuing that experiment, I led the fieldwork at Smithers and visited the Arnold Arboretum of Harvard University to collect tree cores, deepening my expertise in dendrochronological methods.

Relevant activities: Before my current academic path, I earned diplomas in hospitality and sommellerie and spent nearly ten years in those industries, where I built a strong work ethic and sense of rigour. The transition to academia began during six months working a vineyard in New Zealand, where I realized my ties to nature ran deeper than I expected. Thereafter, I enrolled in a Bachelor’s degree in biology, majoring in ecology. During the first year of my MSc, I deepened my quantitative skills through a biomathematics class, a three-week workshop on hierarchical Bayesian models and the Statistical Rethinking course, refining my models in a biweekly Bayesian working group.

My work during my undergraduate studies at Université du Québec à Montréal was recognized with the Governor General’s Academic Silver Medal, the Dean’s List of honours and 13 competitive scholarships totalling \$76,750, alongside a perfect GPA of 4.30/4.30. For my MSc, I received 3 competitive scholarships totalling \$28,750, an honourable mention for my EcoEvo talk, and an Honours designation for my MSc thesis. I have maintained active outreach throughout: I was interviewed at the 2023 UQAM scholarship ceremony, featured by BioChambers for my growth-chamber work, hosted an open house at CSEE (2024) and presented my work twice to the Treepotters community scientists at Harvard (2025–2026).

Across the two years leading my experiment, I trained and supervised 15 undergraduate students in growth-chamber protocols, phenological monitoring and greenhouse management; 2 of them are now pursuing graduate studies. I also led the Smithers field team in 2025. I was a teaching assistant for FRST303 (2024–2025) and FRST304 (2025–2026), running office hours and preparing supplementary materials to help students grasp ecological and plant physiology concepts. Since 2021 I have also mentored 12 international students during their first months in Canada, tutored 7 peers in biochemistry and statistics, and volunteered with Biology Undergraduate Diversity in Research to advise 8 students on applying to graduate school.

My wildlife photography extends this outreach: I won the 2026 Biodiversity Research Center (UBC) photo competition, and my photos have run on the cover of *Le Point Bio* (2024), in the *Proceedings of the National Academy of Sciences* (2025) and in *Le Rappel* (2025).